# ZHENRONG SHEN

Email: zhenrong.shen@manchester.ac.uk | Homepage | Google Scholar | GitHub

## WORK EXPERIENCE

**The University of Manchester (UoM)** — **Manchester, United Kingdom**
*Research Associate; Supervisor: Prof. Alejandro F. Frangi* — *Apr. 2026 – Present*

**ShanghaiTech University** — **Shanghai, China**
*Research Associate; Supervisor: Prof. Qian Wang* — *Jul. 2025 – Mar. 2026*

## EDUCATION

**Shanghai Jiao Tong University (SJTU)** — **Shanghai, China**
*Ph.D. in Biomedical Engineering; Supervisor: Prof. Qian Wang* — *Sep. 2020 – Jun. 2025*

- **Ph.D. Thesis:** Research on Controllable Medical Image Synthesis Methods for Intelligent Disease Diagnosis.

**Beihang University (BUAA)** — **Beijing, China**
*B.Eng. in Biomedical Engineering* — *Sep. 2016 – Jun. 2020*

## RESEARCH EXPERIENCE

### Controllable Lesion Data Synthesis

1. **Chest X-ray Lung Nodule Synthesis for Lung Nodule Detection**
   - Proposed an inpainting-based lung nodule synthesis network and a synthetic sample selection network to select effective synthetic samples for augmenting lung nodule detection. This work was accepted by ***PRCV 2021***.
   - Proposed a lung nodule synthesis framework that disentangles nodule attributes (*i.e.*, shape, size, and texture) and synthesize nodules in a controllable manner, and leveraged such controllability to design a hard example mining strategy for augmenting lung nodule detection. This work was accepted by ***Medical Image Analysis***.
2. **Cervical Cytological Image Synthesis for Cervical Abnormality Screening**
   - Proposed CellGAN, a conditional GAN that generates cytological images of various cervical cell types including NILM, ASC-US, ASC-H, LSIL, and HSIL cells for augmenting patch-level cervical cell classification. This work was accepted by ***MICCAI 2023 (early accept)***.
   - Incorporated CellGAN into a knowledge distillation framework for multi-class cervical cell detection, which facilitates the class-balanced pre-training of a teacher network. This work was accepted by ***Neural Networks***.

### Cross-modality Medical Image Synthesis

1. **PET Image Synthesis for Parkinson's Disease (PD) Diagnosis**
   - Propose a two-stage framework that synthesizes $^{11}$C-CFT PET images from real $^{18}$F-FDG PET scans for multi-modal PD diagnosis, which was based on the correlation between dopaminergic deficiency in the striatum and increased glucose metabolism in PD patients. This work was accepted by ***European Journal of Nuclear Medicine and Molecular Imaging***.
   - Proposed a Metabolism-aware Anomaly Detection (MetaAD) framework, which leverages a cyclic modality translation workflow to identify metabolism anomalies of PD in $^{18}$F-FDG PET scans. This work was early accepted by ***MICCAI 2024 (Young Scientist Award)***.
2. **Whole-body MR-to-CT Synthesis for PET Attenuation Correction**
   - Proposed a whole-body MR-to-CT synthesis framework that integrates structural guidance, spatial alignment, and semantic constraint to enhance synthetic CT image quality, thus facilitating PET attenuation correction in whole-body PET/MR Imaging. This work was accepted by ***Medical Image Analysis***.

## Medical Image Quality Enhancement

**1. Adaptive MRI Motion Artifact Correction**

- Proposed FLEX-MoCo, a flexible MRI motion correction pipeline that effectively identifies motion artifact patterns and dynamically adapts its motion correction strategy. This work was accepted by ***Pattern Recognition***.

**2. MRI Super-resolution for Arbitrary MRI Inter-Slice Spacing Reduction**

- Proposed HiFi-Diff, a diffusion model for arbitrary MRI inter-slice spacing reduction in the through-plane direction by generating in-between MR slices from adjacent MR slices. This work was accepted by ***MLMI 2023***.

## HONORS & AWARDS

- Young Scientist Award, MICCAI 2024 *Oct. 2024*
- Academician Yazhu Chen Scholarship, School of Biomedical Engineering, SJTU *Dec. 2023*
- 2nd Prize in Student Academic Forum, Medical Imaging Computing Seminar (MICS 2023) *Jul. 2023*
- 3rd Prize in Student Academic Forum, Medical Imaging Computing Seminar (MICS 2022) *Jul. 2022*
- 2nd Prize, 17th China Post-graduate Mathematical Contest in Modeling *Dec. 2020*
- Outstanding Graduate Award, BUAA *Jun. 2020*
- 1st Prize, National Biomedical Engineering Innovation Design Competition for College Students *Jul. 2019*
- Meritorious Winner, Interdisciplinary Contest in Modeling (ICM) *Apr. 2019*
- Meritorious Winner, Interdisciplinary Contest in Modeling (ICM) *Apr. 2018*

## JOURNAL PUBLICATIONS

1. **Cross-Modality PET Image Synthesis for Parkinson's Disease Diagnosis: A Leap from [$^{18}$F]FDG to [$^{11}$C]CFT**
   **Zhenrong Shen***, Jing Wang*, Haolin Huang, Jiaying Lu, Jingjie Ge, Honglin Xiong, Ping Wu, Zizhao Ju, Huamei Lin, Yuhua Zhu, Yunhao Yang, Fengtao Liu, Yihui Guan, Kaicong Sun, Qian Wang[#], Chuantao Zuo[#].
   *European Journal of Nuclear Medicine and Molecular Imaging, January 2025*.

2. **Image Synthesis with Disentangled Attributes for Chest X-ray Nodule Augmentation and Detection**
   **Zhenrong Shen**, Xi Ouyang, Bin Xiao, Jie-Zhi Cheng, Dinggang Shen, Qian Wang[#].
   *Medical Image Analysis*, *February 2023*.

3. **Structure-Guided MR-to-CT Synthesis with Spatial and Semantic Alignments for Attenuation Correction of Whole-Body PET/MR Imaging**
   Jiaxu Zheng*, **Zhenrong Shen***, Lichi Zhang[#], Qun Chen.
   *Medical Image Analysis, April 2025*.

4. **FLEX-MoCo: Flexible MRI Motion Correction Using Motion Recognition and Adaptive Routing**
   Feng Li*, **Zhenrong Shen***[#], Jiangdong Cai, Rongrong Xie, Zijian Zhou, Han Zhang, Dinggang Shen, Feng Shi, Qian Wang[#].
   *Pattern Recognition, April 2026.*

5. **Distillation of Multi-class Cervical Lesion Cell Detection via Synthesis-aided Pre-training and Patch-level Feature Alignment**
   Manman Fei, **Zhenrong Shen**, Zhiyun Song, Xin Wang, Maosong Cao, Linlin Yao, Xiangyu Zhao, Qian Wang, Lichi Zhang[#].
   *Neural Networks, October 2024*.

6. **Segment Anything Model for Medical Image Segmentation: Current Applications and Future Directions**
   Yichi Zhang[#], **Zhenrong Shen**, Rushi Jiao.
   *Computers in Biology and Medicine (Highly Cited Paper), March 2024*.

7. **Unleashing the Potential of Generalist Segmentation Foundation Models for Biomedical Image and Video Analysis**

Yichi Zhang, **Zhenrong Shen**, Lanlan Li, Wenbo Zhang, Le Xue[#].
*iRADIOLOGY, April 2026*.

8. **UniCAS: A Foundation Model for Cervical Cytology Screening**
   Haotian Jiang*, Jiangdong Cai*, **Zhenrong Shen**, Mengjie Xu, Manman Fei, Haolin Huang, Xinyu Wang, Rui Bi, Dinggang Shen, Lichi Zhang[#], Qian Wang[#].
   *Cell Reports Medicine, January 2026*.

9. **Improving Self-Supervised Medical Image Pre-Training by Early Alignment with Human Eye Gaze Information**
   Sheng Wang*, Zihao Zhao*, **Zhenrong Shen**, Bin Wang, Qian Wang[#], Dinggang Shen[#].
   *IEEE Transactions on Medical Imaging, January 2025*.

10. **UFFL: Bridging Slide- and Cell-Level Annotations via Unified Feature Fusion Learning for Cervical Cancer Screening**
    Manman Fei, Zhiyun Song, **Zhenrong Shen**, Mengjun Liu, Qian Wang, Lichi Zhang[#].
    *Information Fusion, April 2026*.

11. **Learning Contrast and Content Representations for Synthesizing Magnetic Resonance Image of Arbitrary Contrast**
    Honglin Xiong, Yulin Wang, **Zhenrong Shen**, Kaicong Sun, Yu Fang, Yan Chen, Dinggang Shen, Qian Wang[#].
    *Medical Image Analysis, May 2025*.

12. **Exploring Multi-Connectivity and Subdivision Functions of Brain Network via Heterogeneous Graph Network for Cognitive Disorder Identification**
    Dongdong Chen, Mengjun Liu, **Zhenrong Shen**, Linlin Yao, Xiangyu Zhao, Zhiyun Song, Haolei Yuan, Qian Wang, Lichi Zhang[#].
    *IEEE Transactions on Neural Networks and Learning Systems, October 2024*.

13. **AdLER: Adversarial Training with Label Error Rectification for One-Shot Medical Image Segmentation**
    Xiangyu Zhao, Sheng Wang, Zhiyun Song, **Zhenrong Shen**, Linlin Yao, Haolei Yuan, Qian Wang, Lichi Zhang[#].
    *Expert Systems with Applications, January 2026*.

14. **sTBI-GAN: An Adversarial Learning Approach for Data Synthesis on Traumatic Brain Segmentation**
    Xiangyu Zhao*, Di Zang*, Sheng Wang, **Zhenrong Shen**, Kai Xuan, Zeyu Wei, Zhe Wang, Ruizhe Zheng, Xuehai Wu, Zheren Li, Qian Wang, Zengxin Qi[#], Lichi Zhang[#].
    *Computerized Medical Imaging and Graphics, March 2024*.

15. **A Deep Learning Framework for Synthesizing Longitudinal Infant Brain MRI During Early Development**
    Yu Fang*, Honglin Xiong*, Jiawei Huang, Feihong Liu, **Zhenrong Shen**, Xinyi Cai, Han Zhang, Qian Wang[#].
    *Radiology: Artificial Intelligence, August 2025*.

16. **Uni-COAL: A Unified Framework for Cross-Modality Synthesis and Super-Resolution of MR Images**
    Zhiyun Song, Zengxin Qi, Xin Wang, Xiangyu Zhao, **Zhenrong Shen**, Sheng Wang, Manman Fei, Zhe Wang, Di Zang, Dongdong Chen, Linlin Yao, Qian Wang, Xuehai Wu, Lichi Zhang[#].
    *Expert Systems with Applications, December 2024*.

17. **Spatial Attention-based Implicit Neural Representation for Arbitrary Reduction of MRI Slice Spacing**
    Xin Wang*, Sheng Wang*, Honglin Xiong, Kai Xuan, Zixu Zhuang, Mengjun Liu, **Zhenrong Shen**, Xiangyu Zhao, Lichi Zhang[#], Qian Wang[#].
    *Medical Image Analysis, May 2024*.

18. **Enhancing Generative Models for Modality Imputation of 3-D MRIs via Consistency-Aware Refinement and Super-Resolution Guidance**
    Zhiyun Song, Xin Wang, Honglin Xiong, Mengjun Liu, Manman Fei, Xiangyu Zhao, **Zhenrong Shen**, Sheng Wang, Dongdong Chen, Linlin Yao, Qian Wang, Lichi Zhang#.
    *IEEE Transactions on Neural Networks and Learning Systems, June 2026*.

19. **DW-Net: A Cascaded Convolutional Neural Network for Apical Four-chamber View Ssegmentation in Fetal Echocardiography**
Lu Xu, Mingyuan Liu, **Zhenrong Shen**, Hua Wang, Xiaowei Liu, Xin Wang, Siyu Wang, Tiefeng Li, Shaomei Yu, Min Hou, Jianhua Guo, Jicong Zhang[#], Yihua He[#].
*Computerized Medical Imaging and Graphics, May 2020.*

## CONFERENCE PUBLICATIONS

1. **CellGAN: Conditional Cervical Cell Synthesis for Augmenting Cytopathological Image Classification**
**Zhenrong Shen**, Maosong Cao, Sheng Wang, Lichi Zhang, Qian Wang[#].
*MICCAI 2023 (early accept), October 2023.*

2. **Nodule Synthesis and Selection for Augmenting Chest X-ray Nodule Detection**
**Zhenrong Shen***, Xi Ouyang*, Zhuochen Wang, Yiqiang Zhan, Zhong Xue, Qian Wang[#], Jie-Zhi Cheng, Dinggang Shen.
*Chinese Conference on Pattern Recognition and Computer Vision (PRCV), October 2021.*

3. **MetaAD: Metabolism-Aware Anomaly Detection for Parkinson's Disease in 3D $^{18}$F-FDG PET**
Haolin Huang*, **Zhenrong Shen***, Jing Wang*, Xinyu Wang, Jiaying Lu, Huamei Lin, Jingjie Ge, Chuantao Zuo[#], Qian Wang[#].
*MICCAI 2024 (**Young Scientist Award**), October 2024.*

4. **Arbitrary Reduction of MRI Inter-slice Spacing Using Hierarchical Feature Conditional Diffusion**
Xin Wang*, **Zhenrong Shen***, Zhiyun Song, Sheng Wang, Mengjun Liu, Lichi Zhang, Kai Xuan, Qian Wang[#].
*MICCAI 2023 Workshop on Machine Learning in Medical Imaging (MLMI 2023), October 2023.*

5. **Refining Cervical Cell Classification with Cytological Knowledge and Optimal Attribute Descriptor Matching**
Manman Fei, **Zhenrong Shen**, Mengjun Liu, Zhiyun Song, Yusong Sun, Xu Han, Zelin Liu, Haotian Jiang, Lu Bai, Qian Wang[#], Lichi Zhang[#].
*MICCAI 2025, September 2025.*

6. **MeLo: Low-rank Adaptation Is Better Than Fine-tuning for Medical Image Diagnosis**
Yitao Zhu, **Zhenrong Shen**, Zihao Zhao, Sheng Wang, Xin Wang, Xiangyu Zhao, Dinggang Shen, Qian Wang.
*ISBI 2024 (oral), May 2024.*

7. **One-Shot Traumatic Brain Segmentation with Adversarial Training and Uncertainty Rectification**
Xiangyu Zhao, **Zhenrong Shen**, Dongdong Chen, Sheng Wang, Zixu Zhuang, Qian Wang, Lichi Zhang[#].
*MICCAI 2023 (early accept), October 2023.*

8. **Consistent 3D Medical Image Inpainting via Implicit Representation and Multi-View Sampling**
Yuqi Hu, **Zhenrong Shen**, Zhiyun Song, Dongdong Chen, Liao Wang, Lichi Zhang[#].
*ICICSP 2024, September 2024.*

9. **Weakly Semi-Supervised Cervical Lesion Cell Detection via Twin-Memory Augmented Multiple Instance Learning**
Manman Fei, Zhiyun Song, **Zhenrong Shen**, Mengjun Liu, Qian Wang[#], Lichi Zhang[#].
*MICCAI 2025, September 2025.*

10. **Learnable Subdivision Graph Neural Network for Functional Brain Network Analysis and Interpretable Cognitive Disorder Diagnosis**
Dongdong Chen, Mengjun Liu, **Zhenrong Shen**, Xiangyu Zhao, Qian Wang, Lichi Zhang[#].
*MICCAI 2023, October 2023.*

11. **TracerAD: Training-Free Few-Shot 3D Anomaly Detection for Novel PET Tracers**
Haolin Huang*, Junlei Wu1*, Jiaying Lu, **Zhenrong Shen**, Xinyu Wang, Chuantao Zuo, Qian Wang[#].
*MICCAI 2026 (early accept), September 2026.*

12. **Multi-Tracer Uptake Correction for PET-MR via Aligned-Feature Guidance and Multi-scale Pixel-adaptive Routing**
Aocheng Zhong*, Haolin Huang*, Jing Wang*, **Zhenrong Shen**, Haiyu Song, Yuhua Zhu, Yang Liu, Junlei Wu, Qian Wang#, Chuantao Zuo#.
*MICCAI 2025 (early accept), September 2025*.

13. **Multi-task Screening for Cervical Diseases via Feature Routing and Asymmetric Distillation**
Haotian Jiang*, Haolin Huang*, Jiangdong Cai, **Zhenrong Shen**, Manman Fei, Mengjie Xu, Xinyu Wang, Lichi Zhang#, Qian Wang#.
*MICCAI 2025 (early accept), September 2025*.

14. **RSAD: Region-Specific Anomaly Detection in fMRI for Disease Diagnosis**
Yusong Sun, Dongdong Chen, Mengjun Liu, **Zhenrong Shen**, Zhiyun Song, Yuqi Hu, Manman Fei, Xu Han, Zelin Liu, Xingkai Fang, Lu Bai, Lichi Zhang#.
*MICCAI 2025 (early accept), September 2025*.

15. **Self-supervised Learning with Adaptive Graph Structure and Function Representation for Cross-Dataset Brain Disorder Diagnosis**
Dongdong Chen, Linlin Yao, Mengjun Liu, **Zhenrong Shen**, Yuqi Hu, Zhiyun Song, Qian Wang, Lichi Zhang#.
*MICCAI 2024, October 2024*.

16. **CAS-Net: Cross-View Aligned Segmentation by Graph Representation of Knees**
Zixu Zhuang*, Xin Wang*, Sheng Wang*, **Zhenrong Shen**, Xiangyu Zhao, Mengjun Liu, Zhong Xue, Dinggang Shen, Lichi Zhang#, Qian Wang#.
*MICCAI 2023, October 2023*.

17. **Robust Cervical Abnormal Cell Detection via Distillation from Local-Scale Consistency Refinement**
Manman Fei, Xin Zhang, Maosong Cao, **Zhenrong Shen**, Xiangyu Zhao, Zhiyun Song, Qian Wang, Lichi Zhang#.
*MICCAI 2023, October 2023*.

18. **MITracker: Multi-View Integration for Visual Object Tracking**
Mengjie Xu*, Yitao Zhu*, Haotian Jiang, Jiaming Li, **Zhenrong Shen**, Sheng Wang, Haolin Huang, Xinyu Wang, Han Zhang, Qing Yang, Qian Wang#.
*CVPR 2025 (highlighted), June 2025*.

19. **Gaze-DETR: Using Expert Gaze to Reduce False Positives in Vulvovaginal Candidiasis Screening**
Yan Kong*, Sheng Wang*, Jiangdong Cai, Zihao Zhao, **Zhenrong Shen**, Yonghao Li, Manman Fei, Qian Wang#.
*MICCAI 2024 (early accept), October 2023*.

20. **Longitudinal Image Synthesis for Early Brain Development with Age Awareness**
Yu Fang*, Honglin Xiong*, Jiawei Huang, Xin Wang, **Zhenrong Shen**, Feihong Liu, Xinyi Cai, Han Zhang, Qian Wang#.
*ISBI 2025, April 2025*.

21. **Alias-Free Co-modulated Network for Cross-Modality Synthesis and Super-Resolution of MR Images**
Zhiyun Song, Xin Wang, Xiangyu Zhao, Sheng Wang, **Zhenrong Shen**, Zixu Zhuang, Mengjun Liu, Qian Wang, Lichi Zhang#.
*MICCAI 2023 (early accept), October 2023*.

## PROFESSIONAL ACTIVITIES

- **Journal Reviewer**
*Medical Image Analysis*
*IEEE Transactions on Medical Imaging*
*IEEE Transactions on Neural Networks and Learning Systems*

*IEEE Transactions on Multimedia*
*IEEE Journal of Biomedical and Health Informatics*
*Information Fusion*
*Pattern Recognition*
*Neural Networks*
*NeuroImage*
*Computerized Medical Imaging and Graphics*
*Medical & Biological Engineering & Computing*

- **Conference Reviewer**
  *Medical Image Computing and Computer Assisted Interventions (MICCAI)*
  *IEEE International Symposium on Biomedical Imaging (ISBI)*

- **Teaching Assistant**
  *Computer Vision in Biomedical Engineering (SJTU BME7001)* at Spring 2024